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Department of CSE

B.Tech. CSE Trimester -5

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Generative AI and AI Platform

25CS2207E

WEEK– 5

TOKEN ECONOMICS – COST AND RATE-LIMIT BUDGETING

1. AIM

To develop a Python-based Token Economics Budget Planner to estimate input tokens, output tokens, daily and monthly API costs, and check whether expected application traffic satisfies configured RPM, TPM, and budget limits using a safety margin.

2. APPARATUS / SOFTWARE REQUIREMENTS

1. Computer or Laptop.
2. Python 3.14.
3. PyCharm IDE.
4. Python virtual environment (.venv).
5. Python python-dotenv package.
6. .env configuration file.
7. Python source file (budget.py).
8. Windows PowerShell terminal.

3. OBSERVATION

The experiment uses a Python program to estimate token consumption and API costs before developing an LLM application.

The following parameters are considered:

Parameter	Value
Requests per day	1500
Average input tokens/request	1000
Average output tokens/request	400
Peak requests per minute	25
Input price per million tokens	$0.50
Output price per million tokens	$1.50
RPM limit	50

TPM limit	60000
Daily budget	$2.50
Monthly budget	$60
Safety margin	15%

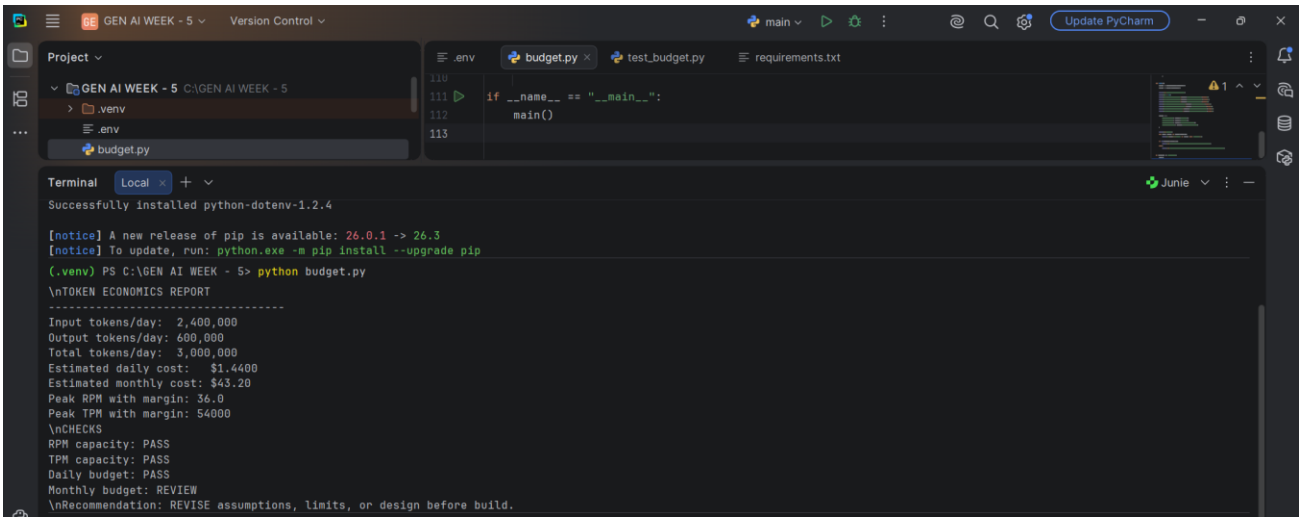
4. FORMULAS

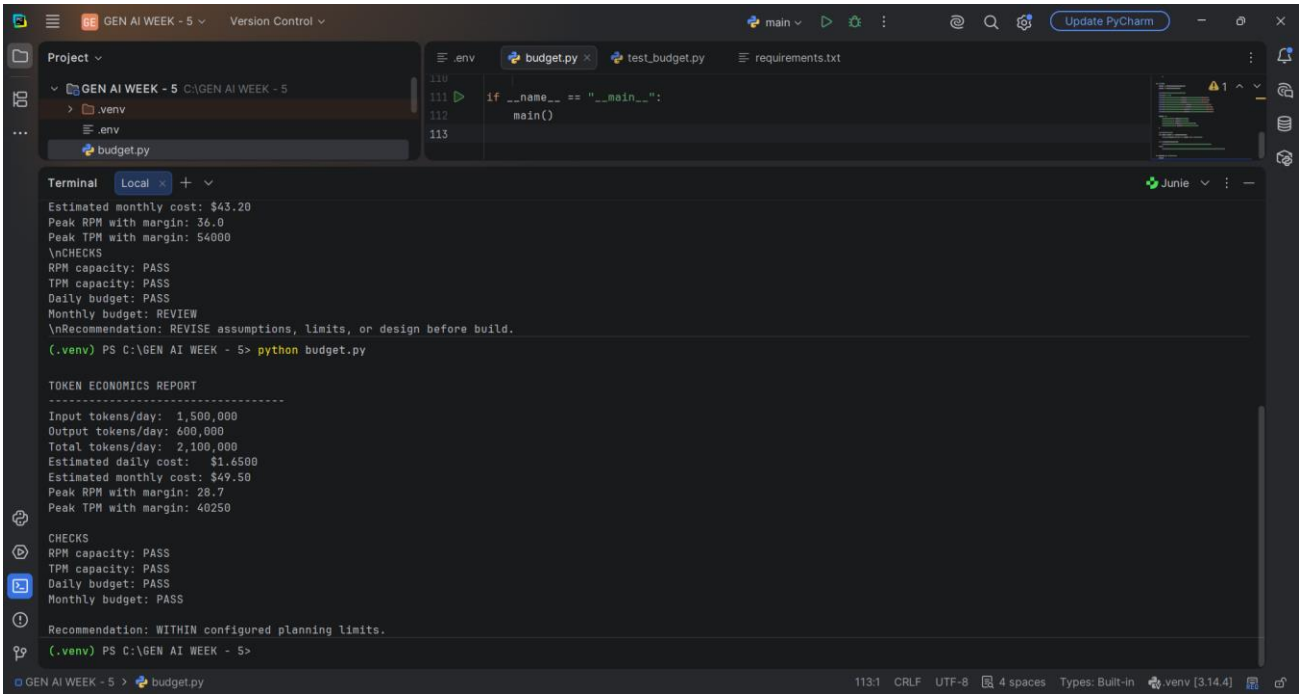
1. Input Tokens per Day = Requests per Day × Average Input Tokens.
2. Output Tokens per Day = Requests per Day × Average Output Tokens.
3. Total Tokens per Day = Input Tokens + Output Tokens.
4. Daily Cost = [(Input Tokens × Input Price) + (Output Tokens × Output Price)] / 1,000,000.
5. Monthly Cost = Daily Cost × 30.
6. Peak RPM with Margin = Peak RPM × (1 + Safety Margin / 100).
7. Peak TPM with Margin = Peak RPM × (Input Tokens + Output Tokens) × (1 + Safety Margin / 100).

5. CALCULATIONS

- Input tokens/day = 1500 × 1000 = 1,500,000.
- Output tokens/day = 1500 × 400 = 600,000.
- Total tokens/day = 1,500,000 + 600,000 = 2,100,000.
- Daily cost = [(1,500,000 × 0.50) + (600,000 × 1.50)] / 1,000,000 = $1.65.
- Monthly cost = $1.65 × 30 = $49.50.
- Peak RPM with 15% margin = 25 × 1.15 = 28.75.
- Peak TPM with 15% margin = 25 × (1000 + 400) × 1.15 = 40,250.

6.OUTPUT





7. RESULT

The Python Token Economics Budget Planner was successfully implemented and executed. It calculated the estimated input tokens, output tokens, total token consumption, daily cost, monthly cost, and peak RPM and TPM values.

All four checks passed:

- RPM capacity: PASS.
- TPM capacity: PASS.
- Daily budget: PASS.
- Monthly budget: PASS.

The final recommendation was **WITHIN CONFIGURED PLANNING LIMITS**.